Analysis of DCO-OFDM and Flip-OFDM for IM/DD Optical-Wireless System

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Abstract— **The paper analyzes the performance of two forms of orthogonal frequency division multiplexing (OFDM) techniques called as DC biased Optical OFDM (DCO-OFDM) and Flip- OFDM for intensity modulated direct detection (IM/DD) system. The aforementioned OFDM schemes are compared in terms of peak-to-average power ratio (PAPR) and bit error rate (BER) as a function of electrical energy per bit to noise power ratio ($E_{b(elec)}/N_0$) when the signal is affected by additive white Gaussian noise (AWGN). In particular, 7-dB and 13-dB DC bias are used for DCO-OFDM in BER analysis and finally compared with flip-OFDM for the different data formats such as 4-,16-,64-,256- QAMs. The results show that flip-OFDM performs better in BER analysis with reduced value of ($E_{b(elec)}/N_0$) and thus, it can be considered as a power efficient unipolar modulation format. However, Flip-OFDM signal provides a high PAPR value that can deteriorate the overall system performance. Finally, OFDM signal performance tradeoff is measured as a function of transmitted optical power, QAM size, and spectral efficiency.**

Index Terms— **DCO-OFDM, Flip-OFDM, PAPR, BER, Intensity Modulation, Direct Detection.**

I. INTRODUCTION

Orthogonal frequency division multiplexing (OFDM) is a fast-progressing multi-carrier modulation scheme that is widely used in most of major wired and wireless communication systems now-a-days [1][2]. Especially in radio frequency (RF) domain, OFDM has been employed as a core part of the physical interface in the communication systems such as worldwide interoperability for microwave access (WiMAX), long term evolution (LTE) mobile network. Moreover, OFDM also has a great potential to use in optical system including optical wireless, multimode and single mode fibre optic communication system [3].

For the cost effective design of OFDM based optical system, the earlier investigations are mainly focused on intensity modulated (IM) optical transmitter with direct detection (DD) optical receiver [4]. For IM/DD system, OFDM signal needs to be unipolar i.e., real and positive, which is quite different from conventional OFDM scheme with bipolar symbols used in RF communication. The real valued OFDM signal is generated by Hermitian symmetric input data with Inverse Fast Fourier Transform (IFFT). However, to generate OFDM signal with positive values, several forms of OFDM schemes have been proposed for IM/DD optical system such as DC-biased optical OFDM (DCO-OFDM), Asymmetrically clipped optical OFDM (ACO-OFDM), flip-OFDM [5][6][7].

In DCO-OFDM, a DC-bias is added to make the OFDM signal positive, where the required bias value depends on the peak-to-average power ratio (PAPR) of OFDM symbols. As the OFDM signal has high PAPR value, the transmit optical power need to be increased, which eventually makes the DCO-OFDM power inefficient. For a power efficient modulation format, ACO-OFDM is proposed to avoid the use of any DC-bias. In ACO-OFDM, only the odd-subcarriers are used to carry the information and clipped the negative values at the transmitter. It is shown that ACO-OFDM is more power efficient than DCO-OFDM, however, in terms of bandwidth utilization, ACO-OFDM is inefficient due to the use of half number of subcarriers for data. Recently, another modulation technique is proposed for unipolar signal called as flip-OFDM, where the positive real time domain signal is transmitted in two consecutive OFDM symbols. It is shown that flip-OFDM provides 50% less implementation complexities compared to ACO-OFDM [5].

This paper provides a comparative analysis between the conventional DCO-OFDM and recently proposed flip-OFDM for IM/DD optical system in terms of bit error rate (BER) performance, PAPR label and optical power efficiency. The BER curves are plotted against electrical energy per bit to noise power ratio assuming that the received signal is corrupted with Additive White Gaussian Noise (AWGN) [8]. The PAPR of DCO- and Flip- OFDM are computed against a range of predetermined threshold values in dB unit. Moreover, the effect of average transmitted optical power on the BER performance is also analysed for both the aforementioned OFDM schemes with respect to normalised bandwidth/bit rate. These results have practical implications in both the wireless and optical fibre IM/DD communication systems.

The rest of the paper is organised as follows. Section II gives a description of system blocks of the two OFDM schemes. Section III provides comparison among those OFDM schemes in terms of BER vs. ($\boldsymbol{E_{b(elec.)}/N_o}$), PAPR and $\mathbf{E_{b(opt.)}/N_0}$ vs. normalised bandwidth / bit rate. Finally, we provide the conclusion in section IV.

II. STUDY OF DIRECT DETECTION OFDM SCHEMES

In this section, we provide brief description about the DCO-OFDM and flip-OFDM systems consecutively.

A. DCO-OFDM

DCO-OFDM transmitter and receiver are shown in Fig.1. At the first step of modulation procedure, a high rate binary

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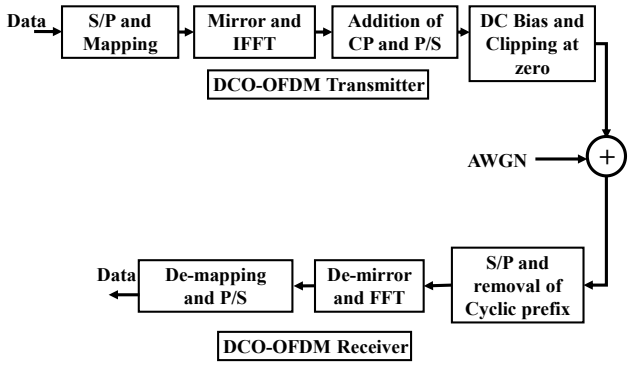


Figure 1: Implementation blocks for DCO-OFDM system

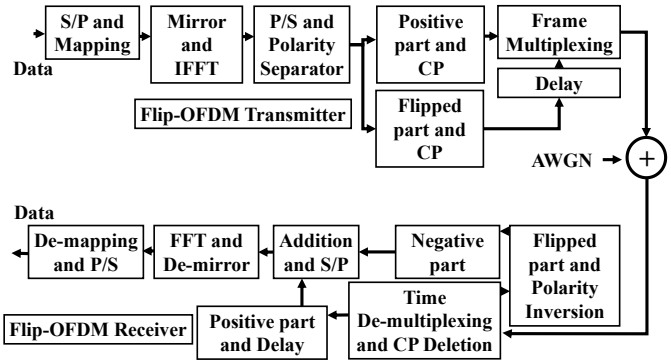


Figure 2: Implementation blocks for Flip-OFDM system

serial data sequence is divided into a number of parallel lower rate data streams that are transmitted simultaneously over a number of subcarriers. Each data stream goes through constellation mapping where every log_2(M) bits are mapped onto complex values according to an *M*-ary quadrature amplitude modulation technique. For $X_n = A_n + jB_n$, where, $n = 0, 1\ldots, N\text{-}1$ denotes the subcarrier number, Hermitian symmetry is taken by complex conjugate mirroring as defined below,

$$X_{N-n} = X_n^* \tag{1}$$

where $n = 0,1,\ldots,N/2$ and $X_0 = X_n = 0$. By taking *N*-point IFFT, real value k^{th} sample is denoted by x_k as,

$$x_k = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} X_n \exp\left(\frac{j2\pi nk}{N}\right) \tag{2}$$

Thus a DCO-OFDM symbol is generated with *N* point real value bipolar data sequence. Afterwards, cyclic prefix (CP) is added in each OFDM symbol and sends to parallel to serial converter for serial data transmission. Finally, DC bias is added and clipped the signal at zero label to ensure non-negative data transmission.

Additive white Gaussian noise (AWGN) is considered in the propagation channel to include noise effect on received data. At the receiver, DCO-OFDM frame is divided into $(N+N_{CP})$ parallel data streams. The number of DCO-OFDM symbols obtained from this serial to parallel conversion process is exactly equal to that of symbols transmitted. Cyclic prefix is then removed from each symbols and demodulation is performed successively using *N* point FFT. The complex valued data streams can be generated from the FFT operation as,

$$X_n = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} x_k \exp\left(\frac{-j2\pi nk}{N}\right) \tag{3}$$

The output of this FFT process is de-mirrored and applied to *M*-ary QAM demodulator to recover the original transmitted binary data streams. At the end, parallel data streams are again converted into serial data streams in binary form and compare with transmit data for BER measurement.

B. Flip-OFDM

Fig.2 shows the block diagram of the generation and reception of Flip-OFDM signal. In the transmitter, the real bipolar OFDM signal is generated following the same procedure like DCO-OFDM signal. However, the incoming bipolar signal is then decomposed into

$$x(k) = x^+(k) + x^-(k) \tag{4}$$

where the positive and negative parts of the Flip-OFDM are defined as,

$$x^+(k) = \begin{cases} x(k) & if\ x(k) \geq 0 \\ 0 & otherwise \end{cases}$$
$$x^-(k) = \begin{cases} x(k) & if\ x(k) \leq 0 \\ 0 & otherwise \end{cases} \tag{5}$$

where $k = 0,1,2,\ldots,N\text{-}1$. Two successive OFDM symbols are used to transmit the aforementioned components. The positive part $x^+(k)$ is transmitted in the first sub-frame, whereas the flipped signal $-x^-(k)$ is transmitted in the second sub-frame. Cyclic prefix is added to both the OFDM frames resulting the length of $(N+N_{CP})$, where N_{CP} is the length of cyclic prefix. Finally, the frames are multiplexed with all the positive real values and send to the desired propagation channel.

At the Flip-OFDM receiver, bipolar OFDM symbols are reconstructed using two data frames namely positive and flipped parts as shown in Fig.2. Firstly, the cyclic prefixes are removed from both of data frames and after this subtraction operation is performed which provides original bipolar signal as,

$$y(k) = y^+(k) - y^-(k) \tag{6}$$

where $y^+(k)$ and $y^-(k)$ represent positive and flipped data frames respectively. FFT operation is performed on this bipolar signal to generate the original complex conjugate data sequences. De-mirroring and QAM demodulation processes are used to reconstruct the binary stream from the complex valued signal.

III. COMPARISON OF DCO-OFDM AND FLIP-OFDM

A. $(E_{b(elec.)}/N_o)$ *vs. BER :*

The $(E_{b\,(elec.)}/N_0)$ vs. BER curves were computed for both type of OFDM signals to compare their performance in AWGN channel. The BER curves of 7 dB biased DCO-OFDM for 4, 16, 64, and 256-QAM are shown in fig.3. To reach BER at 10^{-4}, 600 OFDM symbols are used with 1024

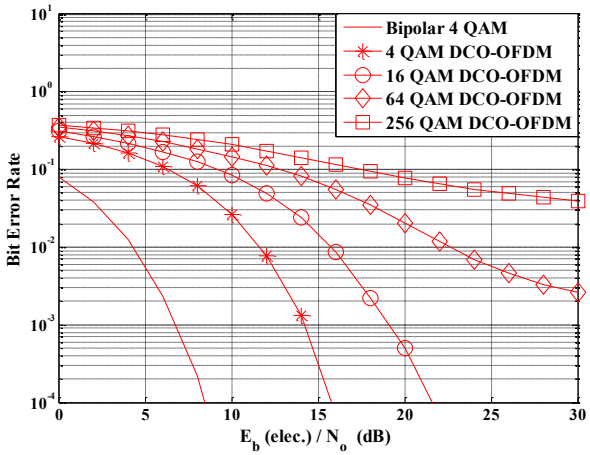


Figure 3: BER analysis for 7 dB biased DCO-OFDM

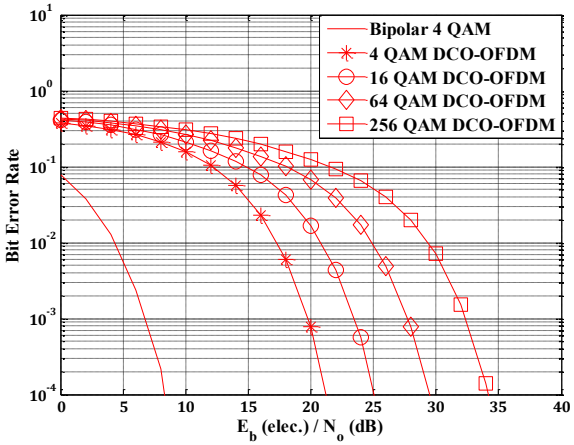


Figure 4: BER analysis for 13 dB biased DCO-OFDM

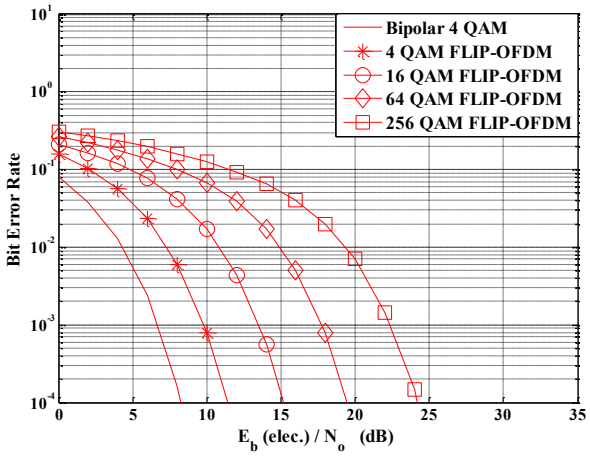


Figure 5: BER analysis for Flip-OFDM

subcarriers in each symbol and the cyclic prefix length was 16. Fig.3 shows that clipping noise dominants for higher constellation size especially in 64- and 256-QAM, which eventually provides the BER floor. However, 4-and 16-QAM modulation formats are still give a better BER performance with the required electrical power higher than the conventional bipolar OFDM. With the increase of DC bias to 13 dB, the clipping noise can be reduced that improves the BER performance for high constellation size modulation formats as shown in Fig. 4. For DCO-OFDM as justified from both the figures 3 and 4, the required $E_b(elec.)/N_0$ for a given BER is approximately equal to the $E_b(elec.)/N_0$ for a bipolar signal plus the bias level in dB. The BER vs $E_b(elec)/N_0$

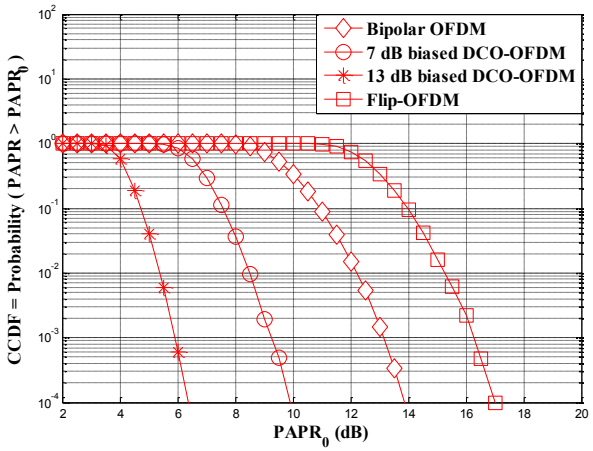


Figure 6: CCDF curves for DCO-OFDM and Flip-OFDM

performance for Flip-OFDM, as shown in fig.5, is more power efficient compare to DCO-OFDM as it requires much less power to achieve the target BER and also to maintain the same data rate. This is mainly due to the average symbol energy which is quite low compare to the DCO-OFDM.

B. *PAPR :*

PAPR is the ratio of peak signal power to average signal power. OFDM signal inherently possess high PAPR as it has large number of subcarriers overlapped to each other. OFDM signal with high PAPR requires a linear power amplifier in the front end transmitter which eventually increases the system implementation cost. Therefore, it is desirable to minimize the PAPR value for the generation of OFDM signal. Fig. 6 shows the PAPR performance for DCO- and Flip OFDM as defined by [9]:

$$PAPR_{dB} = 10 \times \log_{10}\left(\frac{\max[(x_{DCO\,or\,Flip})^2]}{E[(x_{DCO\,or\,Flip})^2]}\right) \qquad (7)$$

The PAPR performance is normally expressed in complementary cumulative distributive function (CCDF). CCDF is defined as the probability that the PAPR exceeds a certain value $PAPR_0$,

$$CCDF = 1 - P(PAPR_{DCO\,or\,Flip}(dB) \le PAPR_0) \qquad (8)$$

For this PAPR measurement, we used an OFDM signal with the IFFT size of 256 and 4-QAM data modulation for each carrier. Fig. 6 shows that Flip-OFDM undergoes PAPR penalty compare to DC biased OFDM signal. In particular, PAPR ratio is around 3.7 dB high than the bipolar OFDM. Moreover, the PAPR value is increased to around 7.2 dB and 10.2 dB for flip-OFDM in compare with 7 dB and 13 dB biased DCO-OFDM. The increase of PAPR for flip-OFDM is mainly due to low value of total average power that arises from many zero values in both positive and flipped frame. However, for DCO-OFDM the large bias voltage improves the average power as well as reduced the negative peaks and thus, 13-dB bias OFDM performs better compare to 7-dB bias for PAPR measurement as shown in fig. 6.

C. *Bandwidth vs.* ($E_{b(opt.)}/N_o$)

We now compare the DCO-OFDM and flip-OFDM in

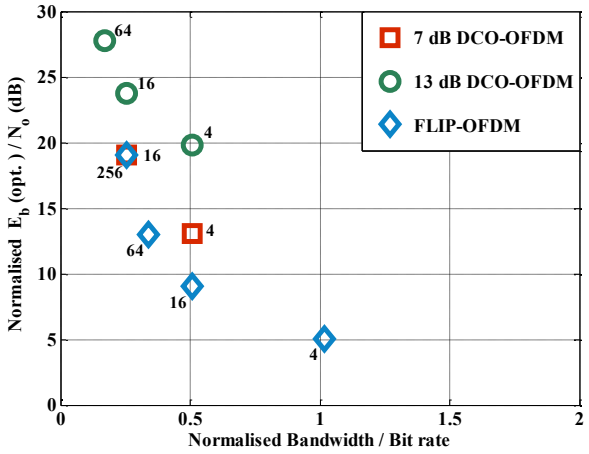


Figure 7: $E_{b(opt.)}/N_0$ for BER 10^{-3} vs Normalized bandwidth per bit rate for DCO- and Flip- OFDM

terms of transmitted optical power to noise power ($E_{b(opt.)}/N_0$) and spectral efficiency as these parameters have the importance in practical implementation. This simulation is performed by adopting unity optical power ($P_{opt.}^{DCO} = P_{opt.}^{FLIP} = 1$) for both Flip and DCO-OFDM. As a result, we can derive the relationship between electrical and optical power for this two OFDM schemes as defined by [7],

$$\frac{E_{b,opt.}^{DCO}}{N_0} = \frac{k^2}{1+k^2}\frac{E_{b,elec.}^{DCO}}{N_0} \quad (9)$$

$$\frac{E_{b,opt.}^{FLIP}}{N_0} = \frac{1}{\pi}\frac{E_{b,elec.}^{FLIP}}{N_0} \quad (10)$$

Here, constant k is obtained by equating the numbers 7 and 13 with the statement $10log_{10}(k^2+1)$ for DCO-OFDM with 7 and 13 dB bias respectively. The simulation results shown in fig.7 clearly reveals that normalised bandwidth per bit rate of 7-dB and 13-dB biased DCO-OFDM with 4-QAM data format are identical and reach to around 0.5. However, Flip-OFDM shows the value of 1.0, which indicates the necessity of double bandwidth for the same modulation format. The reason behind this large bandwidth for Flip-OFDM is the bipolar samples that are divided into positive and flipped parts with equal length each. However, in terms of optical power efficiency, flip-OFDM outperforms the DCO-OFDM as also predicted before. In particular, flip-OFDM shows the power value of 5 dB for 4-QAM data signal, whereas the power increases to 13 dB and 20 dB for the DCO-OFDM with 7-dB and 13-dB bias respectively. With the high constellation size, the performance of flip-OFDM is close to the DCO-OFDM especially with 7 dB bias condition.

IV. Conclusion

In this paper, the performance of Flip-OFDM is compared with DCO-OFDM on the basis of three parameters, BER as a function of ($E_{b(elec.)}/N_o$), PAPR and ($E_{b(opt.)}/N_o$) as a function of normalized bandwidth/bit rate. Flip-OFDM has better BER performance compare to DCO-OFDM, however, it shows high PAPR value mainly due to lower average power. Moreover, flip-OFDM exhibits considerable power gain for the same normalised bandwidth/bit rate with respect to DCO-OFDM. On the other hand, the performance of DCO-OFDM depends on the bias value, where the large bias is useful to reduce the clipping noise.

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